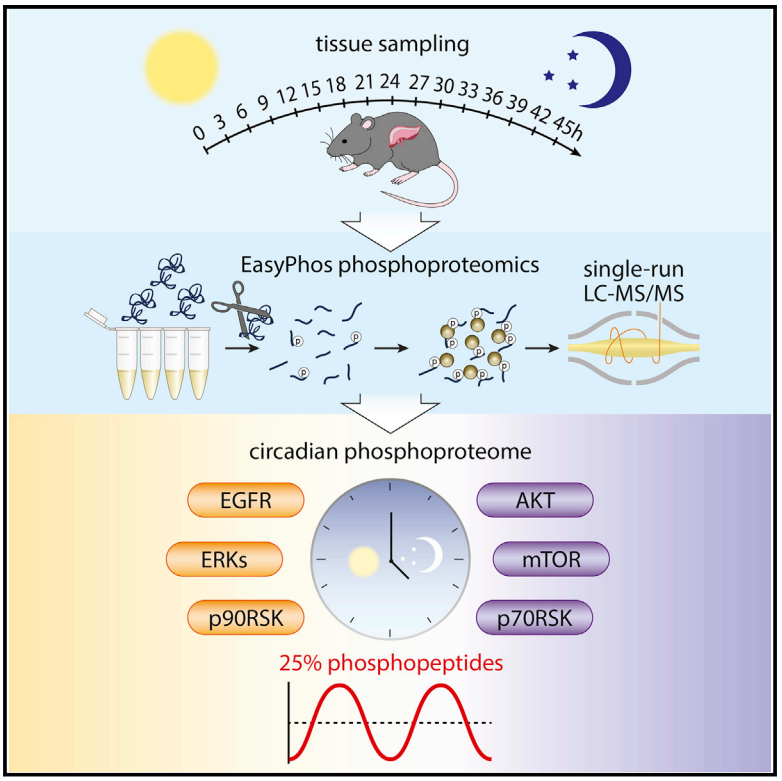
Resource

Cell Metabolism

Phosphorylation Is a Central Mechanism for Circadian Control of Metabolism and Physiology

Graphical Abstract



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In Brief

Robles et al. profile the global in vivo circadian phosphoproteome of the mouse liver and reveal that 25% of the quantified phosphopeptides oscillate with very high amplitudes compared to the transcriptome and proteome. Phosphorylation-dependent tuning of signaling pathways is a key circadian mechanism for metabolic regulation.

Highlights

- Large-scale in vivo circadian phosphoproteome
- More than 25% of sites oscillate daily with large amplitudes in the mouse liver
- Diverse signaling pathways are temporally orchestrated by phosphorylation
- Phosphodynamics mediates crosstalk between circadian clocks and metabolism

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Phosphorylation Is a Central Mechanism for Circadian Control of Metabolism and Physiology

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SUMMARY

Circadian clocks are self-sustainable endogenous oscillators, present in virtually every cell, driving daily cycles of metabolism and physiology. The molecular mechanism of the circadian clock is based on interconnected transcriptional and translational feedback loops. While many studies have addressed circadian rhythms of the transcriptome and, to a lesser extent, the proteome, none have investigated the phosphoproteome. We apply mass spectrometry-based phosphoproteomics to obtain the first global in vivo quantification of circadian phosphorylation in mammals. Of more than 20,000 phosphosites, 25% significantly oscillate in the mouse liver, including novel sites on core clock proteins. The extent and amplitude of phosphorylation cycles far exceeds those observed in RNA and protein abundance. Our data indicate a dominant regulatory role for phosphorylation-dependent circadian tuning of signaling pathways. This allows the organism to integrate different signals and rapidly and economically respond to daily changes in nutrient availability and physiological states.

INTRODUCTION

Temporal organization of nutrient and energy metabolism is critical to maintain homeostasis and, thus, metabolic health. Circadian clocks have evolved to optimize metabolism and physiology to the nutrient fluctuations encountered throughout the day, and almost all metabolic functions at the cellular and systemic levels are under circadian control. Disruption of circadian clocks thus leads to severe metabolic consequences such as diabetes, obesity, and the metabolic syndrome (Lamia et al., 2008; Marcheva et al., 2010; Turek et al., 2005). In mammals, the molecular mechanism of the clock is based on transcriptional/translational feedback loops with the heterodimer transcription factor BMAL1-CLOCK complex at the core. BMAL1-CLOCK binds to consensus DNA sites, E-boxes, found in promoters of target genes, and regulates the expression of their own repressors *Pers* (*per1*, *per2*, and *per3*) and *Crys* (*cry1* and *cry2*), as well as approximately 10% of genes in any given tissue (Panda et al., 2002; Storch et al., 2002; Zhang et al., 2014b). In addition to transcriptional regulation, the circadian clock also orchestrates rhythms of translation, protein abundance (Janich et al., 2015; Mauvoisin et al., 2014; Robles et al., 2014), and metabolites (Dallmann et al., 2012; Eckel-Mahan et al., 2012).

The molecular mechanisms that establish the timing of the clock are based on numerous temporally regulated post-translational modifications (PTMs) of the core clock proteins, which determine their stability, activity, interaction, and cellular localization (Mehra et al., 2009). Protein phosphorylation is the most widely reported PTM of the core clock components and has been shown to be essential for the generation of circadian rhythms in different species (Baker et al., 2009; Reischl and Kramer, 2011). Moreover, non-transcriptional and purely phosphorylation-based circadian cycles with minimal components have been described in cyanobacteria (Nakajima et al., 2005) and *Neurospora* (Larrondo et al., 2015). However, despite being recognized as a key post-transcriptional modification of the molecular clock in mammals, very little is known about the general role of phosphorylation in the circadian regulation of metabolism and physiology. This is mainly because technological challenges have so far precluded the accurate identification and quantification of circadian protein phosphorylation on a global scale. However, recent advances in mass spectrometry (MS) and computational methods now allow the near-comprehensive study and quantification of protein phosphorylation in cells as well as in tissues (Altelaar et al., 2013; Humphrey et al., 2015; Jünger and Aebersold, 2014; Larance and Lamond, 2015; Sharma et al., 2014). Here we adapt these recent developments to generate the first in vivo circadian phosphoproteome. Unexpectedly, we find that a very large part of the phosphoproteome is under circadian control with large fold changes in the mouse liver. This is achieved by activation of a group of kinases leading to temporal integration of signaling pathways. Comparison of the different levels of circadian regulation reveals a crucial role for the phosphoproteome.

RESULTS

Quantifying Global Phosphorylation Cycles in Mouse Liver

To investigate to what extent phosphorylation drives organismal metabolic homeostasis, we measured circadian oscillations of the phosphoproteome in mouse liver. Mice were entrained to light/dark cycles followed by release in constant darkness (see Experimental Procedures) prior to tissue harvesting. Livers were collected from mice sacrificed every 3 hr across 2 consecutive days in biological quadruplicates. Samples were rapidly

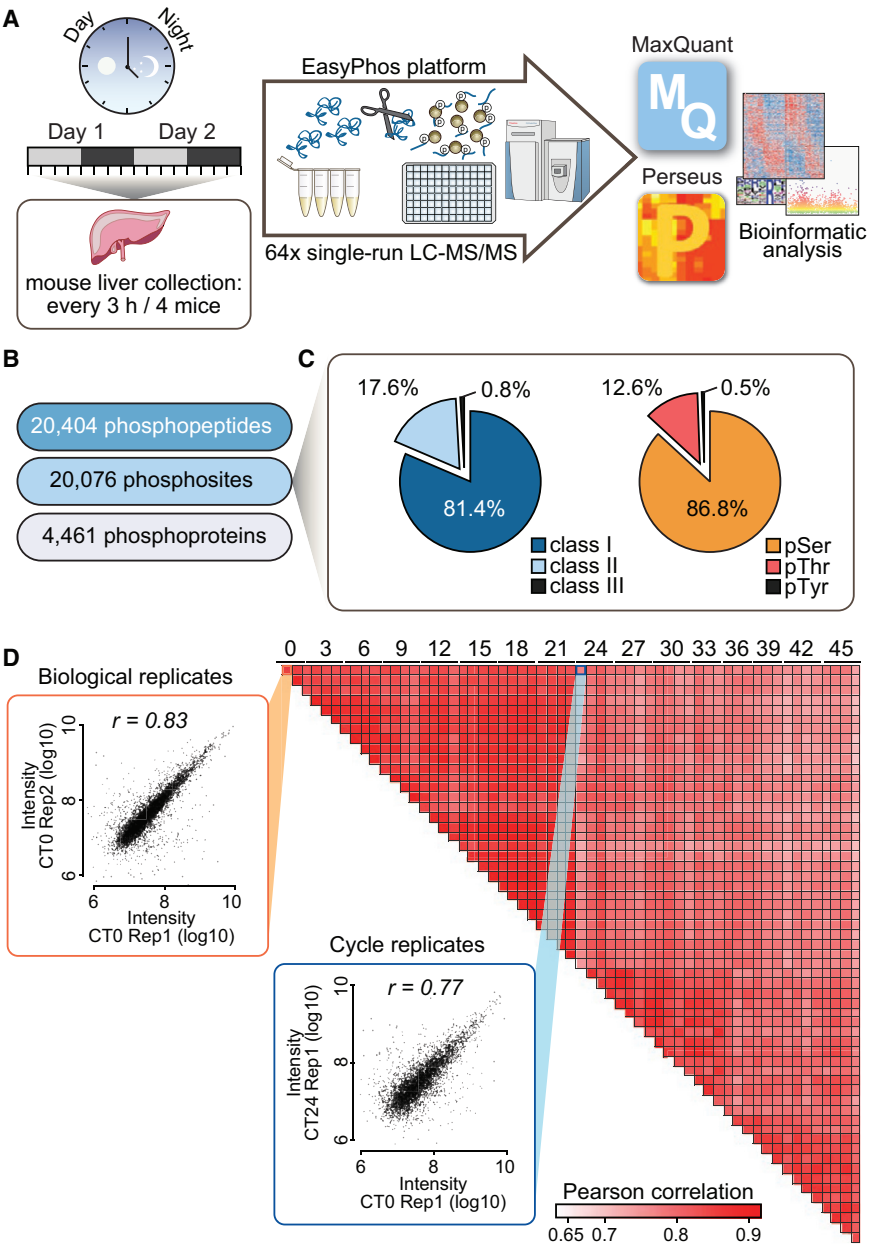


Figure 1. Circadian Phosphoproteome of the Mouse Liver

(A) Samples were collected every 3 hr across 2 days. Phosphopeptides were enriched using the EasyPhos method. Phospho-enriched samples were measured in a single-shot manner using a high-resolution quadrupole-Orbitrap mass spectrometer and processed with MaxQuant following data analysis with the Perseus software package.

(B) Number of phosphopeptides, phosphosites, and phosphoproteins detected in our phosphoproteome of mouse liver.

(C) Distribution of the assigned localization probability (left) and the amino acid residues (right) for all detected phosphosites.

(D) Heatmap of the Pearson correlation coefficients between phosphoproteomes. Representative scatterplots on the right show the correlation between biological replicates (top) and replicates of correlated time points of the two sampled cycles (bottom).

frozen and phosphopeptides enriched from tissue homogenates using the high-throughput EasyPhos platform (Humphrey et al., 2015). This streamlined workflow allowed us to enrich phosphopeptides in a high-throughput manner from the 64 mouse livers prior to single-run measurement in a high-resolution quadrupole-Orbitrap mass spectrometer (Figure 1; Experimental Procedures). Raw MS data were analyzed in the MaxQuant environment for accurate peptide identification, label-free quantification, and phosphosite localization (Sharma et al., 2014). We achieved high phosphoproteome coverage, identifying 20,076 distinct phosphosites on more than 4,400 proteins (Figure 1B), and obtained average phosphorylation site localization probabilities of 0.90% (Figure 1C). These are excellent values, especially considering that we only used 2 mg liver protein lysate and that there was no exogenous stimulation. Measurements showed high reproducibility, with an average phosphopeptide intensity correlation (Pearson r) of 0.85 for all biological replicates and 0.74 between measurements of the first and second cycles (Figure 1D).

More Than a Quarter of the Quantified Liver Phosphoproteome Oscillates Daily

For statistical analysis, we used a stringently filtered dataset of 7,986 phosphopeptides quantified in at least 14 samples across the 2 days (Table S1A, available online; Experimental Procedures). Remarkably, we found that more than 25% of the phosphopeptides, on over 40% of the phosphoproteins, were significantly oscillating in a rhythmic manner in the mouse liver (Figure 2A; Table S2B). We detected oscillations in phosphopeptides across all intensity ranges; thus, cycling phosphopeptide intensities were distributed widely over the five orders of magnitude of all phosphopeptides quantified (Figure S1A) and had excellent phosphorylation site localization probability (average > 98%) (Figure S1B). The proportion of serine, threonine, and tyrosine phosphorylated residues that cycle daily in abundance was similar to that reported by previous large-scale phosphoproteome studies (Figure S1C). Unsupervised temporal clustering showed that the phases of the rhythmic phosphopeptides segregated into two major clusters, corresponding to light phase (when mice are least active) and dark phase (when mice are most active) (Figure 2B). In a principal component analysis (PCA), the corresponding time points from first and second cycle grouped together, and the major component separating the cycling phosphoproteomes was time of the day (Figure 2C).

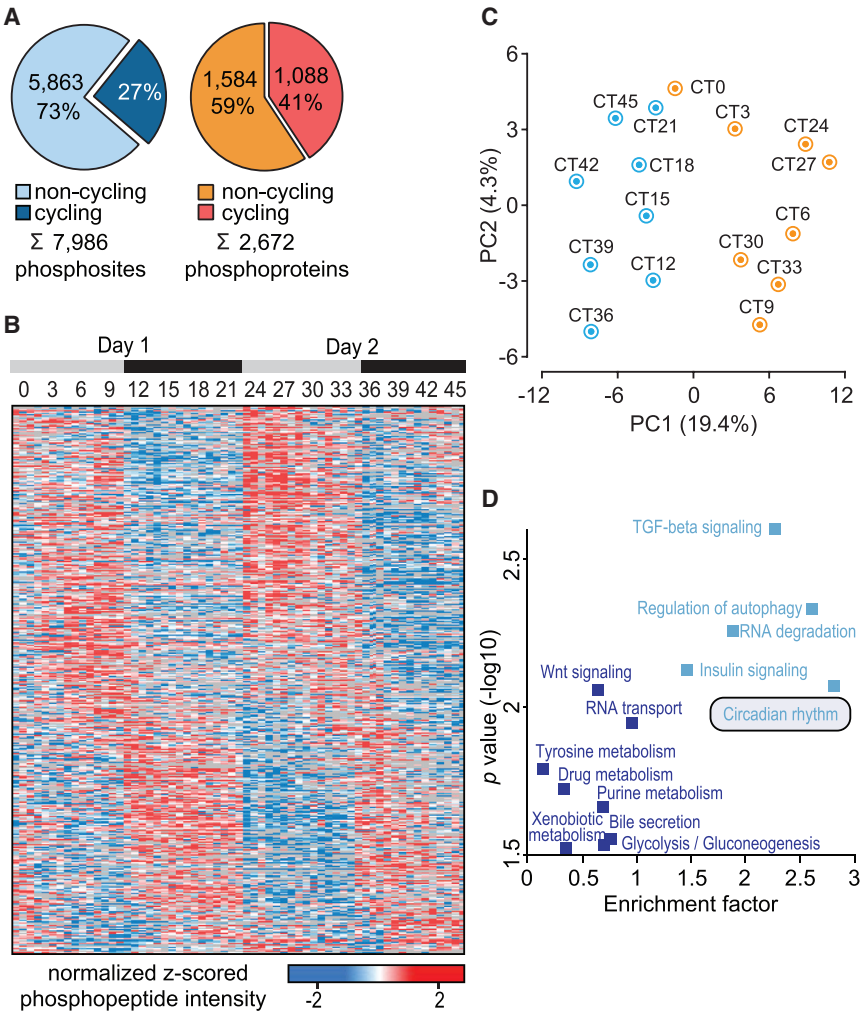


Figure 2. Phosphoproteome Oscillations in the Mouse Liver

(A) Pie chart indicating the percentage of cycling phosphosites (left) and proteins with at least one cycling phosphopeptide (right) in the total quantified liver phosphoproteome.

(B) Hierarchical clustering of the cycling phosphopeptides ordered by the phase of the oscillation. Values for each phosphopeptide (row) at all analyzed samples (columns) are color code based on the intensities, low (blue) and high (red) *Z* scored normalized log10 intensities.

(C) Principal component analysis (PCA) of the cycling phosphoproteome. The major two components separating the data organized the samples into samples corresponding to day (right, orange) and night (left, blue) time points.

(D) Scatterplot shows statistically enriched KEGG pathways in the phosphoproteome ($p < 0.03$). Cycling phosphorylation is found in proteins over-representing the pathways indicated in light blue (enrichment factor > 1), while they under-represent pathways indicated in dark blue (enrichment factor < 1).

We next asked whether cycling phosphosites are predominantly present in proteins belonging to particular signaling pathways and found that the circadian liver phosphoproteome is enriched for proteins from metabolic pathways such as insulin signaling, autophagy, and circadian rhythms (Figure 2D), pointing to a critical role of post-translation in the temporal modulation of these processes.

Comparing the phosphoproteome data to the proteome data measured in the same system before (Robles et al., 2014), we noticed that the phase distribution of rhythmic phosphorylations diverged greatly from the phases of the cycling proteome (Figures 3A and 3B). Strikingly, phosphorylation cycles displayed high amplitudes, with an average of 5-fold change. This is considerably larger than the mean 1.2-fold change observed for protein rhythms (Figure 3C). Moreover, and despite the large amplitudes, the majority (>80%) of the cycling phosphopeptides belong to the smallest quantile (<20%) of the total cumulative phosphopeptide intensity, compatible with regulatory rather than structural roles (Figure 3D). The phases, amplitudes, and extent of dynamic changes in the phosphoproteome indicate that global phosphorylation changes constitute a key circadian post-translational mechanism.

Identification of In Vivo Phosphorylation Cycles in Sites of Clock Proteins

The high enrichment of phosphopeptides from a very complex peptide mixture allowed us to successfully quantify phosphopeptides from low abundant proteins. This even included phosphopeptides from clock proteins, which are typically difficult to detect (Figure 4A). We found cycles of phosphorylation in CRY1 S588 peaking at circadian time 0 (CT0) in total liver, while in nuclei this is reported to be maximal 8 hr later, at CT8 (Gao et al., 2013). Phosphorylation cycles of CRY2 S557 were detected with a peak at the end of the active phase, at CT21. We also identified a novel cycling phosphorylation residue in the clock regulator REV-ERBα, S23, with maximum levels at CT9 preceding the reported maximum binding of this nuclear receptor to promoters of target genes (Bugge et al., 2012). Furthermore, we found in vivo rhythms of CLOCK phosphorylation at the novel sites S446 and S440/S441 (Figure 4B). These residues are located in a serine-rich region of the protein sequence and are highly conserved across mammals (Figure 4C). Interestingly, the peak of the phosphorylated CLOCK peptide coincides with the maximal binding of CLOCK to the promoter of target genes in mouse liver (Koike et al., 2012). Unlike other clock proteins, the levels of CLOCK are constant throughout the day; thus, phosphorylation cycles in S446A and S440/S441 could represent a mechanism to regulate its rhythmic function. To investigate the role of these phosphorylations, we generated a CLOCK mutant construct in which S440, S441, and S446 were substituted for alanine. We then monitored the transcriptional activity of BMAL1 and CLOCK wild-type and triple (S440A/S441A/S446A) mutants of a luciferase reporter containing *Per1* E-boxes. We found that the

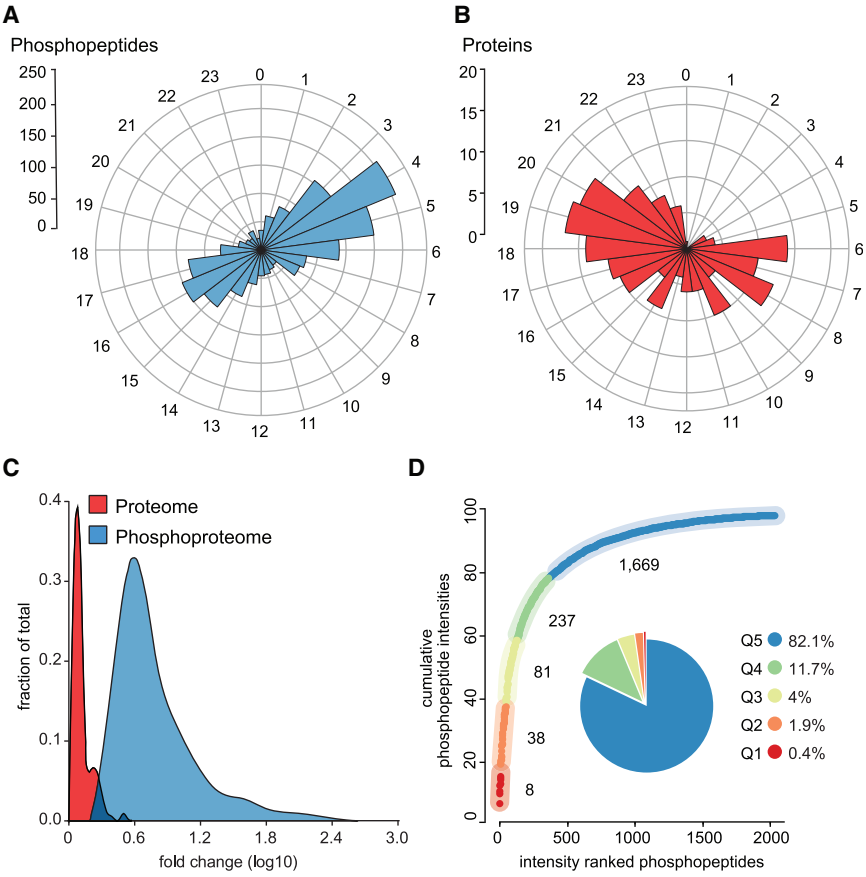


Figure 3. Phase Distribution and Amplitude of the Cycling Phosphoproteome

(A) Rose plots representing the frequency distribution of the phases of the cycling liver phosphoproteome.

(B) Frequency distribution of the phases of the rhythmic proteome (Robles et al., 2014) in mouse liver across the day.

(C) Plot showing the distribution of the amplitudes (fold change of the log10 intensities) calculated for the cycling phosphoproteome (blue) as well as for the rhythmic proteome (Robles et al., 2014) (red).

(D) Scatterplot representing the cumulative intensities of oscillating phosphopeptides ranked ascending. Cumulative intensities were divided into five quantiles and colored as indicated in the figure legend. The pie chart shows the distribution of cycling phosphopeptides in the five defined quantiles.

CLOCK phospho-mutants were transcriptionally less active than wild-type CLOCK (Figure 4D), while CRY1-mediated transcriptional inhibition was not affected (Figure 4E). These data indicate that in vivo phosphorylation cycles at S440/S441 and S446 of CLOCK modulate its transcriptional activity without affecting the CRY1-mediated repression.

Predicting Temporal Activation of Kinases in the Mouse Liver across the Circadian Cycle

Several kinases have been implicated in the molecular mechanism of the clock, but little is known about in vivo temporal activation of kinases with respect to cycles of global phosphorylation. In order to identify key nodes of kinase activity using our in-depth oscillating phosphoproteome, we performed a phase-dependent enrichment analysis using annotated kinase-substrate motifs (see Experimental Procedures). Well-known substrates from a group of kinases clustered throughout the day with specific phases, indicating the temporal activation of these kinases (Figure 5A). Substrates peaking during the inactive phase were enriched with motifs for kinases of the EGFR-MAPK signaling network: ERK, MEK, and RSK, as well as JNK. In the active phase, we observed enrichment of substrates with motifs of kinases downstream of the insulin/IGF1 receptor: AKT, mTOR, and p70S6K kinases, as well as CK1δ, a key component of the molecular clock. Interestingly, the enrichment analysis depicted the phases of kinase activation in the sequential order occurring within the corresponding endogenous signaling cascades: MEK→ERK→RSK and AKT→mTOR→p70S6K. Detailed analysis of functional phosphorylation sites in annotated substrates revealed that ERK substrates showed predominant day phases, while phosphorylation cycles of AKT and mTOR substrates peaked mainly at night (Figures 5 and S2A–S2C). We independently verified phosphorylation cycles of key members of the ERK and mTOR pathway by western blot (Figures S3A and S3B). mTOR-driven phospho-cycles were lost in livers of *Bmal1* knockout mice (Figures S3C–S3E). Together, our data expose an orchestrated network of temporal activated kinases across the day in the mouse liver.

Since the phosphorylation profile of substrates reflects the activity of the target kinase, we inspected our cycling phosphoproteome for uncharacterized substrates of these temporally activated kinases. To this end, we calculated a reference profile from known kinase substrates that oscillate in our dataset, and then extracted those phosphopeptides with the most significantly correlating temporal profiles from the total quantified dataset (permutation-based $q < 0.1$–0.15; see Experimental Procedures). Almost 100 phosphopeptides showed oscillatory profiles resembling the activity of the temporal regulated kinases ERK, AKT, and/or mTOR, making them potential novel in vivo substrates (Figures S2D–S2F; Table S2).

Time-Resolved Map of the Rhythmic Phosphoproteome

The broad spectrum of cellular functions represented by the cycling phosphoproteins points to a wide range of biological processes orchestrated by daily cycles of phosphorylation. To further interpret our dataset, we curated it with data reported in the literature to generate an in vivo time-resolved map of signaling cascades regulated by daily cycles of phosphorylation in the mouse liver. This novel map revealed that globally, endogenous signaling partitioned into two predominant branches, corresponding to the active and inactive parts of the day. The first contained numerous members of the EGFR-RAS-ERK-RSK cascade,

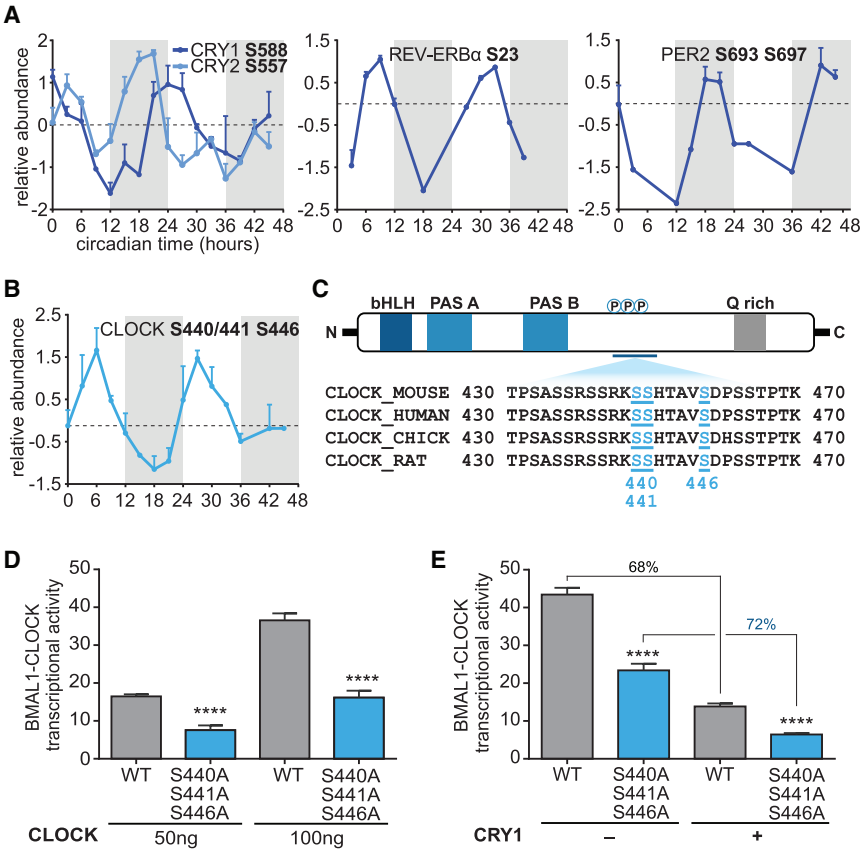


Figure 4. Rhythmic Phosphorylation in Clock Core Proteins

(A) Abundance profiles of phosphopeptides from the clock core proteins CRY1, CRY2, REV-ERBα, and PER2. x axis represents the sampled time points and the y axis the relative abundance (median of *Z* scored log10 intensities for each triplicate ± SEM) of the indicated phosphopeptides. The phosphorylated residues for each protein are indicated in the legend.

(B) Abundance profiles of a CLOCK phosphopeptide with novel modified residues.

(C) Graphical scheme representing the main domains in mouse CLOCK, highlighting the position of the identified cycling phosphorylation. Below, alignment of CLOCK protein sequences from different species containing the identified in vivo phosphorylated residues. The phosphorylated residues, highlighted in color, are conserved.

(D) Transactivation activity of BMAL1 and CLOCK (wild-type in gray and triple S440A/S441A/S446A mutant in blue) of a luciferase reporter from *per1* E-boxes. Bar shows the mean of activity ± SEM (n = 3); ****p < 0.0001 (two-tailed t test).

(E) As in (B) but with the presence of CRY1 to assess CRY1 transcriptional repression of BMAL1 and CLOCK. Numbers on top of the bars indicate the percentage of transactivation CRY1-mediated repression for both conditions. ****p < 0.0001 (two-tailed t test) between wild-type and mutant CLOCK in the absence and presence of CRY1.

concurrently phosphorylated in the inactive phase. In addition, there were oscillating phosphoproteins with phases in the first hours of the day downstream of the kinases GSK3 and AMPK. These included rate-limiting enzymes of glycolysis (PFKFB3), glycogen synthesis (GYS2), and fatty acid biosynthesis (ACACA). Phospho-dependent modulation of these key limiting enzymes during the inactive phase, when the food intake is minimal in mice, leads to stimulation of glycolysis, repression of glycogen synthesis, and inhibition of lipid biosynthesis, respectively.

The second branch comprises numerous oscillating phosphoproteins peaking in the active phase that map to the insulin-AKT-mTOR cascade (Figure 5B). These included AKT itself; members of the two major mTOR complexes, mTORC1 and mTORC2; and several known mTOR substrates. Furthermore, we detected cycles of phosphorylation with predominant night phases among targets of the mTOR effector kinase p70S6K. In the active, high-nutrient phase, therefore, phospho-dependent signals converge on key regulators of metabolic processes with high energy demand such as protein synthesis, growth, and survival. Taken together, our data show that in vivo phosphorylation cycles are directed by nutrient-sensor kinases that ultimately modulate metabolic processes in response to the cellular energy state.

Phosphorylated Signal Integrators at the Intersection between Circadian Clocks and Metabolism

Our time-resolved map revealed a strong interplay between phospho-modulated signaling pathways. This involves direct crosstalk between kinase networks and the assimilation of different temporal phosphorylation signals by relatively few components acting as signal integrators. For instance, our data suggest that the known rhythmic GSK3 phosphorylation with a maximum at the interface of the day and night (Iitaka et al., 2005) is due to AKT activation at the beginning of the night. Likewise, mTOR activity is modulated by the phosphorylation state of its inhibitor, the tuberous sclerosis complex 2 (TSC2) (Huang and Manning, 2008), and we here detected a number of TSC2 residues phosphorylated in a circadian manner, with phases concomitant with the activity of their target kinases, AKT, RSKs, and AMPK (Figures 5B and S4A). Our temporal profiles also show that the phosphorylation of TSC2 by RSK and AMPK during the inactive phase induces its activation, which then leads to the negative regulation of mTORC1 at this time of the day (Kwiatkowski and Manning, 2005). Conversely, we find AKT-mediated phosphorylation of TSC2 S939, which mediates its dissociation to mTORC1, leading to the activation of the complex, peaking at night, concomitant with activation of mTOR. Our data thus indicate that the recognized role of TSC as a metabolic signal integrator (Huang and Manning, 2008) is modulated in a circadian manner, which then temporally orchestrates mTOR-dependent processes, such as protein synthesis and cell survival.

Autophagy is another process downstream of mTOR, and it has been reported to exhibit circadian rhythms in mouse liver, with a peak in the inactive phase (Ma et al., 2011). The mammalian autophagy-initiating kinase ULK1 is a key regulator of this process whose activity is determined by mTOR as well as

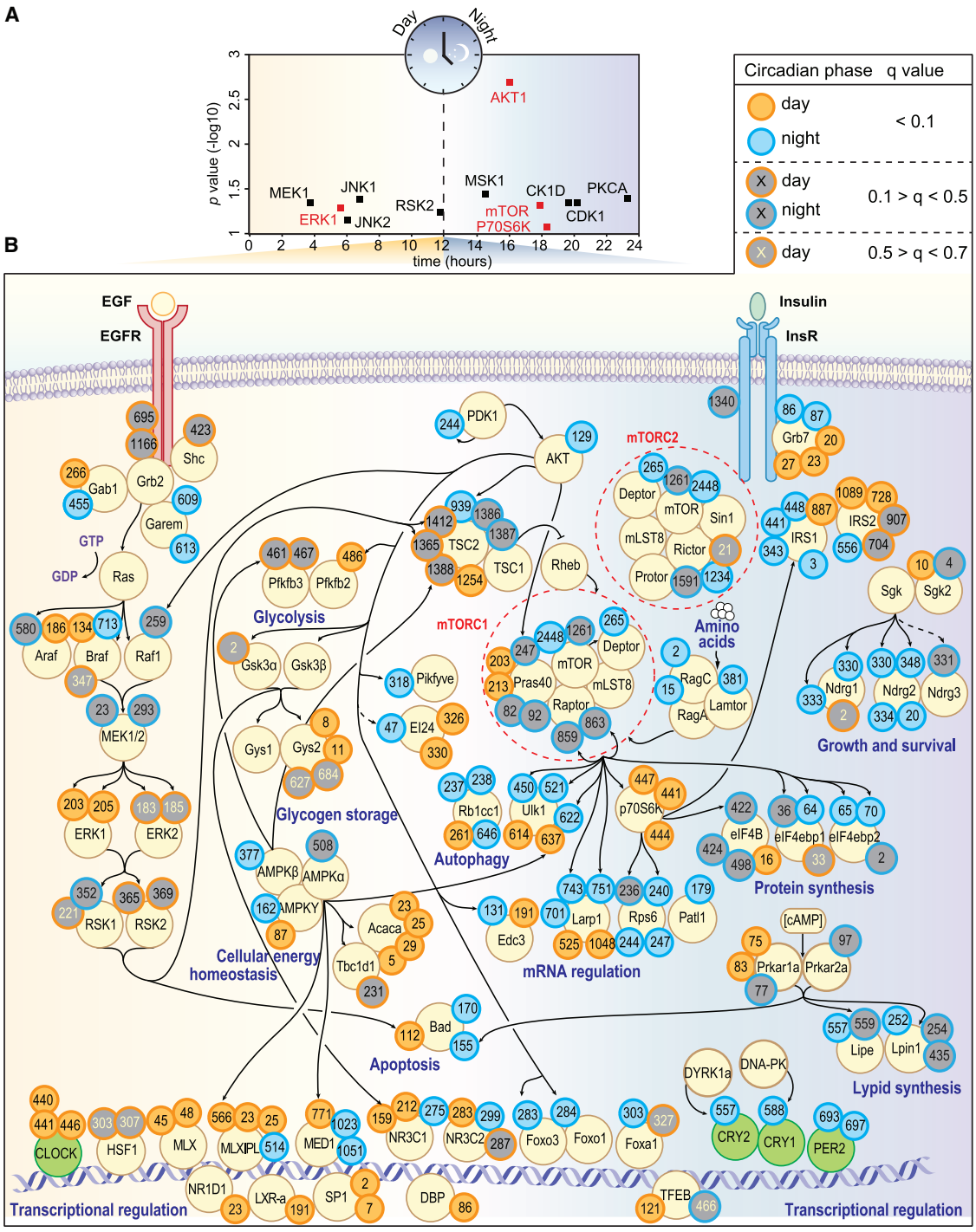


Figure 5. Temporal Regulation of Kinase Networks and Signaling Pathways

(A) Scatterplot showing the result of the kinase annotation enrichment (Fisher's exact text, $p < 0.01$) based on the phase of annotated kinase substrates (PhosphoSite Plus). The x axis indicated the time of the day and the y axis the p value of the enrichment test. Key kinases from the signaling pathways drawn in (B) are highlighted in red.

(B) Time-resolved map of signaling cascades regulated by daily cycles of phosphorylation of their constituents in the mouse liver. Rhythmic phosphorylation sites are shown for each drawn protein color coded based on the phases of the oscillation. The phosphosites are color coded based on the phase of the oscillation; night-peaking phosphosites are blue and day-peaking sites are orange. Blue- and orange-filled phosphosites cycle with statistical significance ($q < 0.1$), while phosphorylation with higher cut-off q values is shown in gray.

AMPK phosphorylation (Kim et al., 2011). Our data show that the distinct temporal activation of AMPK and mTOR in the mouse liver leads to dynamics of ULK1 phosphorylation in several residues with different phases imposing cycles of its activity. Phosphorylation at S450, S521, and S622 matches the activation profile of mTOR kinase, whereas phosphorylation at S614 and S637 follows the cycle of AMPK activation (Figures 5B and S4B). Thus, in the active phase mTOR activation mediates ULK1 inhibition, while AMPK activation, upon low nutrient signal in the inactive phase, promotes ULK1 activity. In this way, our data on the temporal phosphorylation of ULK1 by different kinases could explain the observed cycles of autophagy in the mouse liver.

A crosstalk between EGFR and glucocorticoid-induced signals has recently been reported, involving transcription control through the MAPK pathway (Lauriola et al., 2014). Interestingly, our data revealed the phosphorylation of the EGFR-MAPK pathway constituents to be occurring in anti-phase to the physiological peak of glucocorticoid rhythms, suggesting that phosphorylation contributes to this crosstalk. Moreover, we also identified cycles of phosphorylation with divergent phases in several residues of the glucocorticoid receptor (GR; NR3C1) (Figure 5B). We detected that in vivo cycles of GR S212 and T159 phosphorylation peaked during the inactive phase (Figure S4C), concomitant with its cytoplasmic localization and the activation of its target kinase GSK3 (Ismaili and Garabedian, 2004). In contrast, cycling phosphorylation of GR S284, associated with ligand binding (Chen et al., 2008), peaked during the night, when glucocorticoid levels are high in mice (Figure S4C). The precise orchestration of multi-site phosphorylation of GR exemplifies again how temporally gated phosphorylation enables communication with signal integrators to modulate their function (Figure 5B).

In addition to GR, we detected in vivo cycles of phosphorylation in many other transcriptional regulators. More than 300 rhythmic phosphopeptides matched to 176 proteins with annotated transcriptional regulation activity (Table S3). One-third of the phospho-modulated transcription regulators physically assembled into a protein-protein interaction network (Figure S5; Experimental Procedures). A large cluster of this network comprised core clock components, and this was highly statistically significant (circadian rhythm KEGG [Kyoto Encyclopedia of Genes and Genomes] pathway enrichment; false discovery rate [FDR] = 1.3×10^{-5}). This was also true of several transcription modulators that functionally associate with the molecular clock, such as EP300, CREBBP, and a group of nuclear receptors (FDR = 0.03). Nuclear receptors have been suggested to couple peripheral clocks to metabolic outputs (Zhao et al., 2014). The amplitude and number of phosphorylation sites with clear circadian pattern suggest that phosphorylation could be a critical mechanism to temporally control nuclear receptor activity. In many cases, this may be independent of dynamics at the transcript level, for only two of eight phospho-regulated nuclear receptors (REVERBα and TAK1) have been shown to oscillate at the transcript level in the mouse liver (Yang et al., 2006). In particular, we quantified cycles of phosphorylation on GR, mineralocorticoid receptor (NR3C2), oxysterol receptor LXRα (NR1H3), and the hepatocyte nuclear factor (HNF4α), whereas their transcripts are not cycling, implying post-transcriptional regulation. Taken together, our data show how temporally modulated signaling pathways converge onto the phosphorylation-driven activation of transcription regulators to influence gene expression.

DISCUSSION

Largely for technological reasons, most studies have thus far focused on circadian regulation at the mRNA level, revealing that around 10% of transcripts cycle daily in any given tissue (Panda et al., 2002; Storch et al., 2002; Zhang et al., 2014b). In concordance with this, our recent proteome study in mouse liver found a similar percentage of circadian-regulated proteins, with relatively small amplitude changes (Robles et al., 2014). In stark contrast, our phosphoproteomics data clearly establish that at least a quarter of the phosphoproteome and more than 40% of the phosphoproteins are regulated in a circadian manner in this tissue. Oscillating phosphopeptides show very large amplitude changes compared to the proteome (Figure 3) and transcriptome (Zhang et al., 2014b), and their phases differ greatly from those of rhythmic proteins. One advantage of control at the post-transcriptional level is the rapidity enabled by this mechanism, which can be on the scale of seconds even in a tissue such as mouse liver (Humphrey et al., 2015). This is in contrast to many minutes or hours required to enact gene expression changes. Therefore, while it is clear that cells and tissues can modulate circadian functions at the level of gene, protein, and/or PTMs, our results suggest that the major locus of control is at the level of PTMs such as phosphorylation, many of which will eventually impinge upon gene and protein outputs. This mode of regulation is exquisitely economical, since it takes about 1,200 ATPs to produce a protein of an average size of 300 aa (Milo et al., 2010); however, only a single ATP is consumed by the addition of a phospho-group. Thus, the maintenance of cellular protein phosphorylation levels is estimated to consume <0.02% of the total cellular energy consumption (Shacter et al., 1984). The intensity of >80% of all the cycling phosphopeptides comprises less than 20% of cumulative phosphopeptide intensities, further indicating the limited pool of ATP required for circadian control by phosphorylation (Figure 3D).

This study provides a first glimpse at how many molecular switches are employed to adapt tissues to the different metabolic demands that they encounter throughout the day. We found more than 2,000 significantly changing phosphosites on more than 1,000 proteins, which represents a lower limit because advances in phosphoproteomics technologies will increase this number further. Even allowing for some level of bystander phosphorylation events among this tightly regulated set, it appears likely that the number of signaling nodes influenced by the circadian clock is in the thousands. It is already known that non-transcriptional feedback loops based entirely on phosphorylation can generate circadian oscillations in *Neurospora* and cyanobacteria (Baker et al., 2009; Johnson et al., 2008) and that the mammalian core clock proteins are controlled in part by phosphorylation (Reischl and Kramer, 2011). Based on the results presented here, we propose that phosphorylation is also the central mechanism regulating cellular physiology across the day. Dynamic kinase-substrate

relationships are ideally suited to integrate temporal signals between general metabolic programs and the core clock. How critical an intact clock is for the temporal integration of phosphorylation is shown by the lack of AKT activation in response to feeding or insulin stimulation in mice lacking hepatic BMAL1 (Zhang et al., 2014a). Supporting this notion, we observed that phosphorylation cycles of the mTOR pathway discovered here are lost in livers from *Bmal1*$^{-/-}$ mice (Figure S3).

Expanding from a few known examples like AMPK (Jordan and Lamia, 2013), our data reveal a plethora of dynamic phosphorylated substrates of temporally activated kinases that may serve as critical nodes in the crosstalk between circadian and metabolic cycles. Of these, a few proteins such as TSC2, ULK1, and GR work as central integrators of temporal phospho-dependent signals from different kinase networks.

Moreover, a fraction of the rhythmic phosphoproteins are targets of widely prescribed drugs such as nuclear receptors. Hence, further investigation of the specific and systemic effects of cycling phosphorylation reported here will greatly contribute to improvements in chronotherapy as well as the understanding of crosstalk between metabolism and circadian clocks.

EXPERIMENTAL PROCEDURES

Animals and Tissue Collection

Eight-week-old female and male C57BL/6J mice were housed in light-tight boxes with free access to food and water and entrained to a 12 hr/12 hr light-dark schedule for 10 days before being transferred to complete darkness. After 1 day in constant darkness, mice were sacrificed at 3 hr intervals over 2 days (Robles et al., 2014). Prior to liver excision, mice were perfused with ice-cold PBS to remove blood content. Livers were then quickly frozen in liquid nitrogen followed by storage at −80°C. All mice were bred and maintained in the animal facility of the Max Planck Institute of Biochemistry according to institutional guidelines, and all animal experiments were approved by the government agencies of Oberbayern.

Phosphoproteome Sample Preparation

Protein extracts from mouse livers were obtained by homogenization of 1 mg liver per 1 mL lysis buffer (0.1 mM Tris-HCl [pH 7.6], 4% SDS, and 0.05 mM DTT supplemented with complete protease and phosphatase inhibitor cocktails [Roche]) using an Ultra Turbax blender (IKA) at maximum speed at 4°C for 30–60 s. Homogenates were then sonicated in a bioruptor (4°C for 15 min, or until a homogeneous suspension was formed). After 5 min of incubation at 95°C, the mixture was cleared by centrifugation at 16,000 *g* at room temperature for 10 min. Protein content was determined by tryptophan fluorescence as previously described (Robles et al., 2014). A total of 2 mg protein lysates were precipitated with acetone, and phosphopeptides were enriched using EasyPhos (Humphrey et al., 2015). As described previously, pellets were resuspended in 500 μL TFE digestion buffer and protein digestion was performed by adding 1:100 (protein:enzyme) trypsin and LysC with rapid agitation (2,000 rpm) overnight at 37°C. To the digested peptides, 150 μL 3.2 M KCl, 55 μL 150 mM KH2PO4, 800 μL acetonitrile (ACN), and 95 μL trifluoroacetic acid (TFA) were added. Peptides were mixed at room temperature for 1 min at 2,000 rpm, cleared by centrifugation, and transferred to clean 2 mL 96-well deep-well plates. TiO_2 beads were subsequently added to peptides at a ratio of 10:1 beads/protein, suspended in 80% ACN/6% TFA, and incubated at 40°C for 5 min at 2,000 rpm. Beads were subsequently pelleted by centrifugation for 1 min at 3,500 *g*, and the supernatant (containing non-phosphopeptides) was aspirated and discarded. Beads were suspended in Wash buffer (60% ACN and 1% TFA), transferred to clean 2 mL 96-well-plates, and washed a further four times with 1 mL wash buffer. After the final wash, beads were suspended in 100 μL transfer buffer (80% ACN and 0.5% acetic acid), transferred onto the top of a C8 StageTip, and centrifuged for 3–5 min at 500 *g* or until no liquid remained on the StageTip. Bound phosphopeptides were eluted two times with 30 μL Elution buffer (40% ACN and 15% NH_4OH [25%, HPLC grade]) and collected by centrifugation into clean PCR tubes. Peptide samples were concentrated in a SpeedVac for 15 min at 45°C and acidified by the addition of 10 μL 10% TFA. Phosphopeptides were desalted by loading onto styrenedivinylbenzene-reversed-phase sulfonated (SDB-RPS; 3M Empore) StageTips in 100 μL SDB-RPS wash buffer (0.2% TFA) and centrifuged for 3 min at 500 *g*. StageTips were washed with 100 μL wash buffer, and phosphopeptides were eluted with 60 μL elution buffer (80% ACN and 5% NH_4OH [25%, HPLC grade]) by centrifugation for ~5 min at 500 *g*. Samples were immediately concentrated in a SpeedVac for 30 min at 45°C to a final volume of 2 μL. Peptides were resuspended in buffer containing 2% ACN and 0.1% TFA for liquid chromatography-tandem mass spectrometry (LC-MS/MS) analysis.

LC-MS/MS Analysis and Data Processing

Prior to MS, peptide mixtures were separated on a 50 cm reversed-phase column (diameter of 75 μm packed in-house with ReproSil-Pur C18-AQ 1.9 μm resin [Dr. Maisch GmbH]) over a 270 min gradient of 5%–50% buffer B (0.1% formic acid and 80% ACN) using the EASY-nLC 1000 system (Thermo Fisher Scientific) with a flow rate of 300 nL/min. The nLC system was coupled to a Q Exactive HF mass spectrometer (Thermo Fisher Scientific), acquiring full scans (300–1,600 m/z, R = 60,000 at 200 m/z) at a target of 3e6 ions. The ten most intense ions were isolated and fragmented with higher-energy collisional dissociation (HCD) (target 1e5 ions, maximum injection time 120 ms, isolation window 1.4 m/z, NCE 27%, and underfill ratio of 20%) and detected in the Orbitrap (R = 15,000 at 200 m/z).

Raw MS files were processed within the MaxQuant environment (version 1.5.3.2) using the integrated Andromeda search engine with FDR < 0.01 at the protein, peptide, and modification level. The search included variable modifications for oxidized methionine (M), acetylation (protein N-term), and phospho (STY) and fixed modifications for carbamidomethyl (C). Peptides with at least six amino acids were considered for identification, and "match between runs" was enabled with a matching time window of 0.7 min to transfer MS1 identifications between runs. Peptides and proteins were identified using a UniProt FASTA database from mouse (September 2014) containing 51,210 entries.

Data are available via ProteomeXchange with identifier ProteomeXchange: PXD003751.

Bioinformatic and Statistical Analysis

Bioinformatic analysis was performed with the Perseus software (Tyanova et al., 2016). Briefly, label-free intensities in every sample were normalized by subtracting the median of all the intensities in each sample. Phosphopeptides without label-free intensities in >14 samples were then filtered out. We did not consider replicate number 4 from time point CT0, CT33, and CT39, since the number of identifications in these samples was very limited. For statistical cycling analysis, we fit the label-free phosphopeptide intensities to a cosine with a fixed period of 23.6 hr and with amplitude and phase as free parameters (Robles et al., 2014). Profiles were ranked by their variance ratio. This was the part of the variance explained by the fit divided by the contribution to the variance that was not accounted for by the fit. Based on this ranking, we determined a permutation-based FDR by repeating the same procedure 1,000 times on the same profiles but with randomized time labels. We used a statistical cut-off of $q < 0.1$ to define the cycling phosphoproteome. Hierarchical clustering was performed in a phase-preserving manner, by restricting the order of elements to that determined by the output of the cosine model-based fitting. We used protein annotations from UniProtKB, Gene Ontology (GO), KEGG, and PhosphoSitePlus. Enrichment analysis of KEGG pathways was performed using Fisher's exact test. We extracted transcription regulators by filtering by the Uniprot keywords "transcription," "transcription regulation," and "transcription elongation." Phase-dependent annotation enrichment analysis was performed using a cut-off for cycling phosphopeptides of $q < 0.33$. The same dataset (cycling $q < 0.33$) was used for the extraction of a maximum of 30 phosphopeptides with the most similar profile (correlation profiling, permutation-based $q < 0.1$ for ERK and AKT, $q < 0.15$ for mTOR) to a reference profile generated with known annotated kinase substrate from PhosphoSitePlus (Hornbeck et al., 2015). Kinase-substrate networks were

annotated using information from PhosphoSitePlus and visualized using Cytoscape 3.2.1. Protein interaction networks were generated with the STRING database (version 10.0) (Szklarczyk et al., 2015) and visualized with Cytoscape 3.2.1. The PCA was performed in the Perseus software as described (Robles et al., 2014). Briefly, the total dataset was used as the initial matrix, in which missing values were imputed by drawing random numbers from a normal distribution shifted to simulate signals from low abundant proteins. The width parameter of this normal distribution was chosen as 0.3 of the SD of all measured values and the center was shifted toward low abundance by 1.8 times this SD. A cycling analysis test (period = 23.6 hr) was then performed, and only sites with $q < 0.1$ were used for the PCA. The PCA was performed by singular value decomposition.

Transactivation Assays

HEK293 cells were transfected with FuGene HD (Promega), and transactivation assays were performed essentially as described (Robles et al., 2010), using 100 ng pcDNA3-Bmal1, 50 or 100 ng pcDNA3-Clock, and 10 ng pcDNA3-Cry1 expression plasmid per well (Invitrogen); 22 ng pGL3-Per1-E54bp firefly luciferase reporter (Promega); and 7.5 ng pRL-CMV (Renilla) luciferase internal control plasmid (Promega).

Western Blotting and Antibodies

Liver homogenizes were loaded onto 4%–20% Bis-Tris NUPAGE gels (Thermo), and after electrophoresis, proteins were transferred to nitrocellulose and blocked in 10% BSA on TBS with 0.1% Tween-20 for 1 hr. Incubation of primary antibodies was done overnight at 4°C in 5% BSA on TBS with 0.1% Tween-20. Antibodies were obtained from Cell Signaling: mTOR pS448 (2971), RPS6 p235/p236 (2211), RPS6 p240/p244 (2215), p90RSK pS221 (3556), EGFR pT669 (3056), ERK1/ERK2 pT202/Y204 (8544), BAD pS221 (5284), and GAPDH (2118). Densitometry quantification was performed with ImageJ software.

ACCESSION NUMBERS

The ProteomeXchange accession number for the phosphoproteome data reported in this paper is ProteomeXchange: PXD003751.

SUPPLEMENTAL INFORMATION

Supplemental Information includes five figures and three tables and can be found with this article online at http://dx.doi.org/10.1016/j.cmet.2016.10.004.

AUTHOR CONTRIBUTIONS

Conceptualization, M.S.R. and M.M.; Methodology, M.S.R. and S.J.H.; Investigation, M.S.R.; Writing – Original Draft, M.S.R., S.J.H., and M.M.; Writing – Reviewing & Editing, M.S.R., S.J.H., and M.M.; Funding Acquisition, M.S.R. and M.M.; Supervision, M.S.R. and M.M.

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